Physics Based Simulation of a Three Dimensional LiDAR Sensor

MAE 5810 Robot Perception

Midterm Project Report

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Abstract

This project models a spinning three dimensional pulsed LiDAR with opaque visibility, surface response, attenuation, and receiver noise. A cluttered navigation task supplies 22 controlled settings, 66 closed-loop runs, and 66 replays at identical baseline poses. A direct 3D diagram separates field of view from occlusion. An independent 33,000-trial experiment probes detection near the receiver threshold at three navigation viewpoints; its probability calibration RMS error is 0.55 percentage points. A further 720 runs compare 12 sensor conditions across three fixed layouts and 20 noise seeds. Run-level confidence intervals, collision and timeout counts, and failure-aware clearance distributions expose task limits. Measured maps show why successful navigation does not imply complete perception. These experiments verify the virtual model and its dependencies within stated assumptions; physical calibration remains future work.

1 Introduction

A virtual LiDAR must determine where its beams can reach and whether their returned signal is measurable. A missing point may arise from angular sampling, occlusion, a range gate, or weak power. This project therefore models measurements together with field of view (FOV) and line of sight (LOS), addressing the dependencies on sensor state, targets, and environment required by the midterm assignment [1].

The sensor uses eight channels for the navigation studies. Physical surface modeling is motivated by Heiden et al. [2], whose continuous wave phase estimator differs from this pulsed geometric approximation. Controlled replays isolate measurement effects, threshold trials resolve uncertain detections, and repeated navigation tests measure consequences. Target geometry interventions are interpreted together with their visibility changes.

2 Theory and Methods

2.1 Scan pattern

The sensor frame has x forward, y left, and z up. World z is up. With M azimuth samples per full revolution and L channels, beam angles and the nominal unit direction are

$$\alpha_i = \alpha_{min} + \frac{2\pi i}{M}, \qquad \beta_j = \beta_{min} + \frac{j(\beta_{max} - \beta_{min})}{L - 1} \tag{1}$$

$$b_{ij} = [\cos \beta_j \cos \alpha_i, \cos \beta_j \sin \alpha_i, \sin \beta_j]^T \tag{2}$$

A full sweep uses i = 0, …, M − 1. A partial window uses the same azimuth spacing. A single channel takes the configured upper elevation. Angles in equations are in radians; plots use degrees.

2.2 Pose and coverage

All channels at one azimuth share a firing time. The origin translates at constant world velocity; yaw may change during a scan while roll and pitch remain fixed:

$$t_i = \frac{\alpha_i - \alpha_{min}}{2\pi f_{rot}}, \qquad o_i = o_0 + v t_i \tag{3}$$

The sensor to world rotation is R = Rz(yaw) Ry(pitch) Rx(roll), and the world beam is d = Rb. For point p, let q = R^T(p − o), r = ‖q‖, a = atan2(qy, qx), and e = arcsin(qz/r). The nominal FOV is

$$F = \{ p : r_{min} \le r \le r_{max},\ a \in A,\ \beta_{min} \le e \le \beta_{max} \} \tag{4}$$

A is the circular azimuth window. Discrete beams can miss targets within this continuous set. For each subray, the nearest opaque intersection with scene surfaces S is

$$r_k = min \{ s > 10^{-6}\ m : o + s d_k \in S \} \tag{5}$$

The first hit is found before range gating, so a weak or too near object still blocks farther geometry. A finite beam uses normalized Gaussian weighted subrays in the plane normal to its central direction. For full divergence δ, the angular standard deviation is δ/4; a sunflower pattern truncated at 2.5 standard deviations supplies the sample offsets.

2.3 Surface power and echo formation

For a ray facing normal n, μ = max(0, −n^Td). Diffuse weight ρd, specular weight ρs, and exponent m define the approximate surface law

$$f = \frac{\rho_d}{\pi} + \frac{\rho_s (m + 2)}{2\pi} [max(2\mu^2 - 1,\ 0)]^m \tag{6}$$

$$P_k = w_k P_t A_r \eta\ f \frac{\mu_k}{r_k^2}\ \exp(-2\kappa r_k) \tag{7}$$

Here Ar = π(Dr/2)², η is optical efficiency, and κ = κair + arain q_{rain}^b. Invalid hits contribute zero power. Rain and fog change attenuation, without backscatter returns. Consecutive subray hits separated by at most the pulse gate merge into an echo group G:

$$\Delta r_{gate} = \frac{c \tau_{FWHM}}{2}, \qquad P_G = \Sigma P_k, \qquad \bar{r}_G = \frac{\Sigma P_k r_k}{\Sigma P_k} \qquad (k \in G) \tag{8}$$

The centroid approximates pulsed ranging rather than waveform processing; it can lie between surfaces at edges. Geometry supports planes, spheres, yawed boxes, and capped cylinders.

2.4 Detection and range uncertainty

For an existing echo, noisy power is $\tilde{P} = PG + \nu$, with $\nu \sim N(0, \sigma P^2)$. Detection requires $\tilde{P} \geq \gamma\sigma P$, giving the single echo probability

$$P_D(G) = \Phi\left(\frac{P_G}{\sigma_P} - \gamma\right) \tag{9}$$

Φ is the standard normal cumulative distribution function. The strongest mode selects the greatest noisy detected power; first and last modes select by range. There is at most one retained return per nominal beam. With $S = \tilde{P}/\sigma P$, the assumed timing inspired range law is

$$\sigma_r^2(S) = \sigma_0^2 + \frac{(c\,t_{rise}/2)^2}{2\ max(S,\ 10^{-9})}, \qquad z = \bar{r}_G + \sigma_r(S)\xi \tag{10}$$

ξ is independent standard normal noise. This law is an assumption, not a derived Cramér–Rao bound. Selection truncates measured power but leaves the subsequent standardized range noise normal. Coordinates use the central beam and known firing pose:

$$\hat{p}^S = z\ b, \qquad \hat{p}^W = o + z\ d \tag{11}$$

2.5 Mathematical dependencies

A forward plane intersection is $r = n^T(p0 - o)/(n^Td)$. Horizontal ground at mounting height h and downward world beam elevation β therefore gives

$$r = -\frac{h}{\sin\ \beta}, \qquad \frac{\partial r}{\partial h} = -\frac{1}{\sin\ \beta}, \qquad \frac{\partial r}{\partial \beta} = \frac{h\ \cos\ \beta}{\sin^2\ \beta} \tag{12}$$

Height and orientation change range, with increasing sensitivity near horizontal beams. The FOV transformation changes coverage with pose; the nearest hit rule changes LOS at occluder edges. For a beam filling diffuse target, $P = C \exp(-2\kappa r)/r^2$ and $C = PtAr\eta\rho d \cos\theta/\pi$, so

$$\frac{\partial\ \ln\ P}{\partial r} = -\frac{2}{r} - 2\kappa, \qquad \frac{\partial\ \ln\ P}{\partial \kappa} = -2r, \qquad \frac{\partial\ \ln\ P}{\partial \theta} = -\tan\ \theta \tag{13}$$

Diffuse weight scales power linearly. Greater range, extinction, and incidence angle weaken the diffuse signal, and therefore reduce detection probability. The angle derivative holds range fixed and excludes the specular branch. For any selected physical reference,

$$E[(z - r_{ref})^2 \mid \tilde{P},\ G] = (\bar{r}_G - r_{ref})^2 + \sigma_r^2(S) \tag{14}$$

The first term is centroid displacement; the second is conditional random variance. Angular, pose, calibration, and additional physical model errors are excluded.

3 Procedure

Python and NumPy generate measurements; independent renderers read saved results. The primary cluttered task has 12 objects: nine in the planning map and three unknown to the planner. Its designated target is a pure diffuse box centered at (11, 1.1, 0.6) m, sized (0.8, 1.6, 1.2) m, with diffuse weight 0.30.

The baseline uses eight channels from -20 to +10 degrees, 360 azimuths, one subray, a level 0.6 m mount, and 0.5-200 m range gates. Parameters are 75 W peak power, 25 mm aperture, 0.5 efficiency, 4×10^{-9} W power noise RMS, threshold factor 10, 0.01 m range floor, 5 ns pulse, and 1 ns rise time.

A* supplies a reference from (0, 0) to (18, 0) m on a 0.25 m grid, with 0.90 m total inflation and checked shortcuts. The measured-point controller uses $0.12 \le z \le 1$ m, without target labels or true obstacle geometry. Limits are 1.2 m/s, 90 degrees/s, 0.2 s steps, and 36 s duration. Forward Euler integrates

$$p_{k+1} = p_k + v_k \Delta t \, [\cos \theta_k, \sin \theta_k]^T, \qquad \theta_{k+1} = wrap(\theta_k + \omega_k \Delta t) \tag{15}$$

3.1 Controlled dependencies

Vary pose, FOV, LOS, target geometry and reflectance, attenuation, noise, sampling, and enabled sensing in 22 settings. Each has three closed-loop runs and three replays at 271 common baseline snapshots (seeds 1-3). Count emitted target intersections, nearest visible hits, in-gate eligible echoes NE, and detections ND separately. For qi from Equation (9),

$$\hat{p}_D = \frac{N_D}{N_E}, \qquad \bar{p}_D = \frac{\Sigma q_i}{N_E} \qquad (i \in E) \tag{16}$$

3.2 Threshold and robustness studies

At three saved approach poses, extract one eligible target ray at ranges 7.99, 4.90, and 1.93 m. Fix each ray and solve extinction to obtain mean SNR 6-14 at 11 settings. Draw 1,000 actual scans per setting using 33,000 disjoint noise seeds, independent of navigation. Keep all misses in the denominator. These coefficients are illustrative optical stress, not calibrated weather.

Freeze 12 conditions on the original three environment presets, including barrels, distinct from the primary box-target scene. Run 20 seeds (1001-1020) per cell: 720 runs. Reuse each layout's partial-map reference and controller. Test extinction 0.0002, 0.1, 0.3, 0.6, 1.2, 3, and 10 m^{-1}, high mount, sparse sampling, front and side FOV, and disabled sensing. Retain every failure; one whole run is the statistical unit. Paired seeds do not imply identical per-ray noise.

4 Results

4.1 Sensor pose and visibility

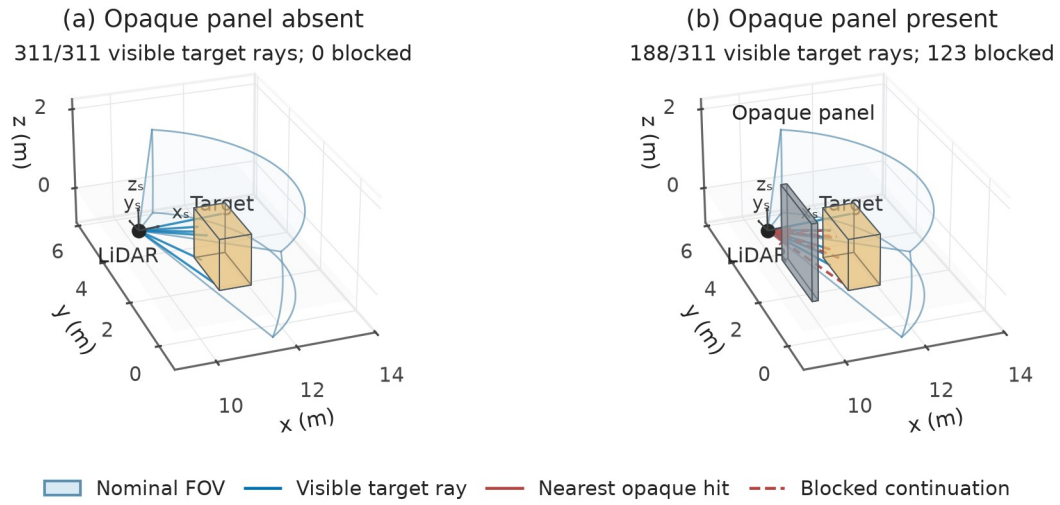


Figure 1 Same-pose 3D coverage and occlusion at 8.2 s. Blue rays reach the target; solid red rays stop at the panel; dashed red segments show blocked continuations. Scene geometry explains visibility; it is not a measured cloud.

The angular FOV is ±60 degrees horizontally and -20 to +10 vertically. Display ends at 4 m; physical gates are 0.5-200 m, with the near cutout omitted. Each panel emits 968 beams, including 311 target intersections. The panel changes 311 visible target hits into 188 visible and 123 blocked. All rays are audited; representative rays are drawn. The envelope contains channel gaps.

Height changes the intersections predicted by Equation (12). In the primary replays, raising the mount from 0.6 to 1.6 m reduces eligible target echoes from 46,712 to 21,621 and worst navigation gap from 0.330 to 0.096 m. Pitch offsets also move the sampled surfaces; positive pitch points downward.

Post-hoc geometry at the open high-mount run's last four poses (seed 1001) shows 152-173 visible barrel rays per pose, none in the controller's height band. Restoring a 0.6 m mount at identical XY/yaw gives 286-544 in-band intersections. Visible 3D surfaces can therefore miss body-relevant coverage. High mounting causes 20/20 collisions in open and narrow layouts; cluttered 19/20 goals accompany a median gap of only 1.8 mm.

Full-circle yaw of 60 degrees preserves geometry by permuting the grid. Primary replays retain 14,421 target opportunities with front 120-degree FOV; sideways coverage has none and causes three collisions. Its conditional detection probability is undefined. The LOS panel also changes physical obstacle geometry.

4.2 Target and environmental dependencies

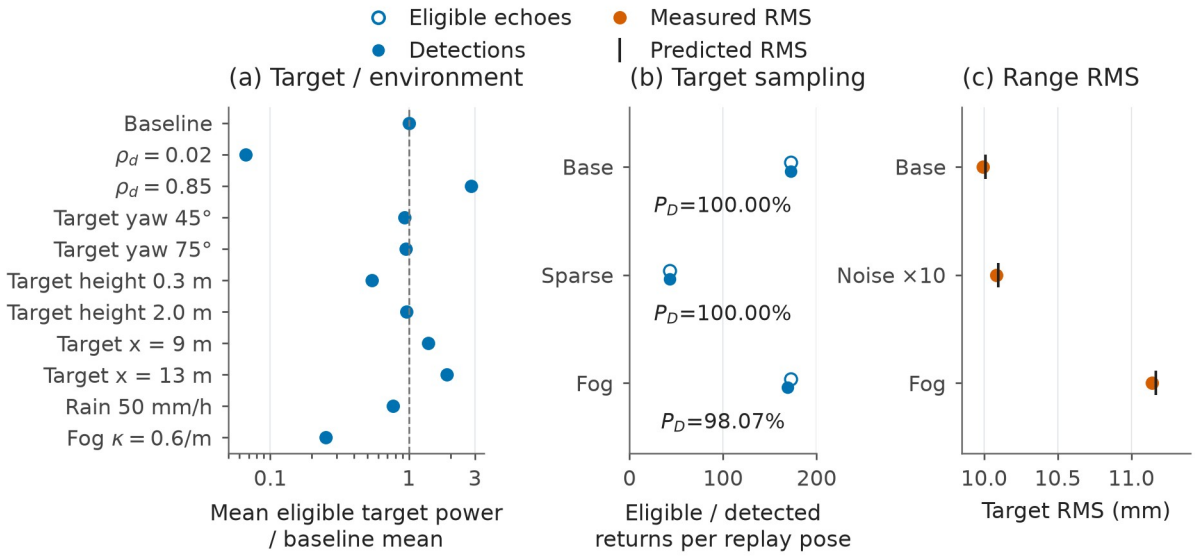


Figure 2 Fixed-pose replay comparisons. Power ratios use independently evaluated noiseless eligible-beam predictions; counts and range errors use measured returns. Moved, rotated, and resized targets change geometric exposure. Range RMS is conditional on detected target returns.

Diffuse weights 0.02, 0.30, and 0.85 give power ratios 0.0667, 1, and 2.8333, following Equation (7). All eligible target echoes remain detected and paths remain unchanged: a physical power change need not alter a strongly illuminated target's detection or the task.

Target heights of 0.3, 1.2, and 2.0 m give 17,851, 47,562, and 50,191 intersections. This exposes discrete angular sampling of target size. Target yaw 45 and 75 degrees changes mean eligible incidence to 36.16 and 34.74 degrees. Rotation also changes projected area and footprint, so these cases demonstrate combined geometry and surface response, rather than a pure cosine-only intervention.

Target position is compared at one common viewpoint. At seed-1 time 6 s, moving the target center to x = 9, 11, and 13 m gives mean eligible ranges 2.134, 4.011, and 5.950 m and powers 0.339830, 0.105144, and 0.048696 mW. The decrease is consistent with range and incidence in Equation (7). Pooled whole-route power can reverse this ordering because the target exposures and range gates differ.

Rain at 50 mm/h and extinction 0.6 m^{-1} reduce mean eligible power from 0.745081 to 0.575356 and 0.187126 mW, consistent with Equation (13). Four-degree azimuth spacing reduces eligible target echoes from 46,712 to 11,592. Sampling removes opportunities; attenuation weakens echoes.

Tenfold receiver noise changes detected-target range RMS from 9.992 to 10.082 mm, versus predictions 10.010 and 10.095 mm. The 10 mm floor dominates these strong echoes; the threshold study below probes weaker signals.

4.3 Near-threshold detection and range uncertainty

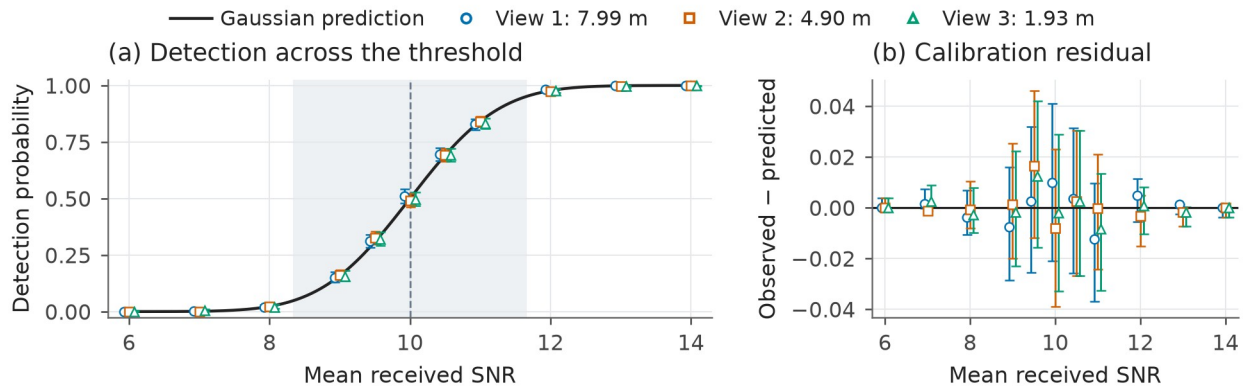


Figure 3 Detection and calibration residuals from 33,000 actual scan trials. Each marker counts 1,000 trials, including misses; bars are pointwise 95% Wilson intervals. Small horizontal offsets distinguish viewpoints. The shaded band is the predicted 5-95% transition, not an uncertainty interval.

The original replays nearly saturate detection. Here extinction changes the mean SNR of three fixed eligible navigation rays; geometry, reflectance, receiver noise, and threshold remain fixed. Ray selection does not depend on noisy detection.

At mean SNR 10, predicted probability is 0.5; measured fractions are 0.510, 0.492, and 0.498. Over 33 settings, calibration RMS error is 0.547 percentage points, with maximum residual 1.646 points. The model lies within 32/33 pointwise intervals; the excluded prediction is retained.

With n independent trials, observed fraction $\hat{p}$, and a = 1.96, the Wilson interval is

$$\frac{\hat{p} + \frac{a^2}{2n} \pm a\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{a^2}{4n^2}}}{1 + \frac{a^2}{n}} \tag{17}$$

For detected echoes only, define range and standardized errors; misses are not zero errors:

$$e_i = z_i - r_{true,i}, \qquad \varepsilon_i = \frac{e_i}{\sigma_r(S_i)}, \qquad RMSE = \sqrt{\frac{\Sigma e_i^2}{N_D}} \tag{18}$$

Across 16,507 detections, range RMS is 32.358 mm versus 32.287 mm predicted from measured-SNR conditional variance; standardized RMS is 1.0018. Agreement checks the assumed noise and selection laws, not physical accuracy. Intervals do not provide simultaneous coverage.

4.4 Repeated navigation and failure boundaries

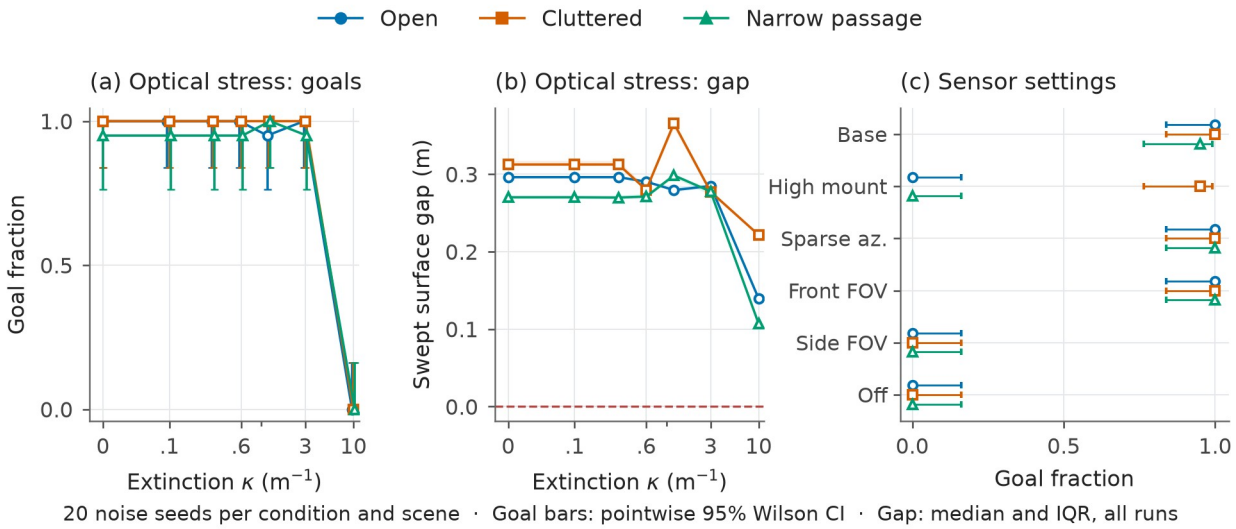


Figure 4 Frozen 720-run study. Goal bars: pointwise 95% Wilson intervals. Gap: all-run median and interquartile range. Sensor settings: 1.6 m mount, 90 azimuths, 120-degree FOV. Extinction is illustrative optical stress, not calibrated weather.

Table 1 Goals/collisions/timeouts; each cell totals 20. All 36 groups are retained in the accompanying results.

Condition	Open G/C/T	Cluttered G/C/T	Narrow G/C/T
Baseline	20/0/0	20/0/0	19/0/1
$\kappa = 10\ m^{-1}$	0/0/20	0/0/20	0/0/20
Mount 1.6 m	0/20/0	19/1/0	0/20/0
Side FOV 120°	0/20/0	0/20/0	0/20/0
LiDAR disabled	0/20/0	0/20/0	0/20/0

Baseline reaches 59/60 goals. Wilson intervals remain [0.839, 1.000] for 20/20 and [0.000, 0.161] for 0/20; one whole run is the unit. These pointwise intervals apply to fixed layouts and this controller, not new random environments. Side FOV rotates the matched front window by 90 degrees.

At $\kappa = 10\ m^{-1}$, the open representative stalls from 8.6 to 36 s despite nearby returns. Open-layout outcomes at $\kappa = 1.2$ (19/20) and $3\ m^{-1}$ (20/20) have overlapping intervals, without establishing monotonic task reliability. Time/path summaries condition on success; paired time differences require both successes. Clearance retains failures. Poses are exact, objects static, and scans snapshots.

4.5 Route generation and measured 3D mapping

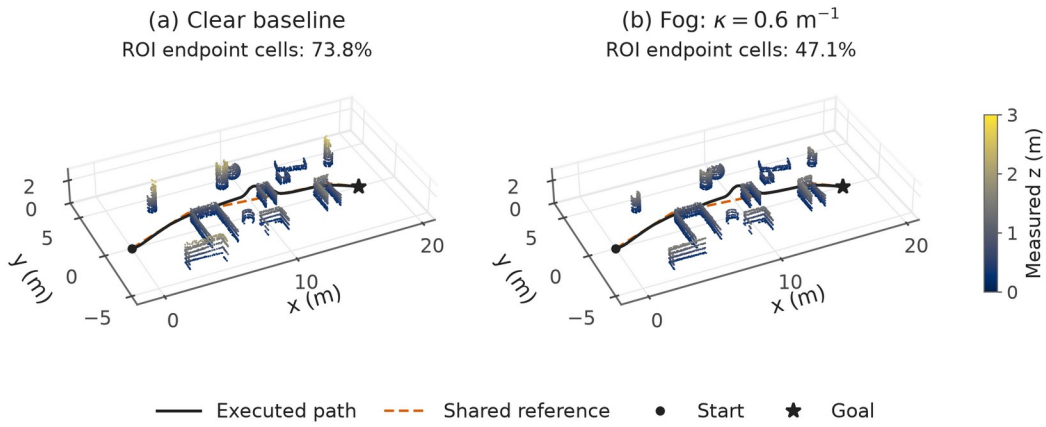


Figure 5 Seed-1 measured maps: black driven path, orange shared reference, height colors. Display hides z ≤ 0.12 m. Each map ends with its run; no scene truth is drawn.

Measured surfaces accumulate in 0.12 m voxels. Endpoint cells are 0.2 m over [-2, 21] m × [-6, 7] m (NROI = 7,475). For K intervals and K + 1 states,

$$L = \Sigma \| p_{k+1} - p_k \|, \qquad f_{obs} = \frac{N_{obs}}{N_{ROI}}, \qquad k = 0, \ldots, K - 1 \tag{19}$$

Primary-task attenuation reduces mean observed cells from 73.7 to 47.1 percent despite three successful runs. Endpoint counts do not establish accuracy or free space. Collision uses swept gap ≤ 0; goal tolerance is 0.35 m.

Independent numerical checks additionally cover 65 geometry and 72 power cases, with maximum residuals 3.83e-13 m and 4.39e-16 relative power.

5 Conclusions

Controlled replays expose LiDAR dependencies; threshold trials verify probabilistic behavior; 720 navigation runs reveal collisions and timeouts. Goal completion does not imply complete perception. Results support the virtual model within its assumptions. Physical calibration, backscatter, pose uncertainty, navigation scan motion, and latency remain future work.

References

[1] “Midterm project,” Robot Perception, course assignment handout, pp. 1-2.

[2] E. Heiden, Z. Liu, R. K. Ramachandran, and G. S. Sukhatme, “Physics-based Simulation of Continuous-Wave LIDAR for Localization, Calibration and Tracking,” Proc. IEEE ICRA, 2020, pp. 2595-2601.